Bar Brain: Real-Time Barbell Rep Analysis Using Accelerometer Signal Processing

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Abstract

Weightlifting is one of the most popular forms of exercise worldwide, yet lifters at every level lack affordable, reliable tools to objectively measure their performance. Most rely on a training partner's input, a phone propped against a wall, or simply how a set "felt" to judge the quality of their repetitions. These solutions lack the quantifiable data needed to track progress or catch form breakdowns before they lead to injury.

Metrics such as range of motion, pause time at the top and bottom of reps, and barbell speed have significant implications for both the effectiveness and safety of barbell training. Bar Brain is a sensor and software suite designed to bridge this gap by collecting accelerometer data directly from the barbell and transforming it into reviewable performance metrics.

The suite uses an Arduino Nano 33 BLE housed in a 3D-printable enclosure that affixes onto any standard barbell, streaming motion data over Bluetooth to a companion web application where each set is automatically broken into individual reps and analyzed for range of motion, velocity, acceleration, and duration. Bar Brain eliminates reliance on camera-based estimation or demanding price tags with a low-cost, sensor-driven solution that gives weightlifting enthusiasts the objective feedback they need to train smarter and safer.

System Architecture

Sensor Layer (C++)
Arduino Nano 33 BLE Rev2 + BMI270 IMU
Madgwick filter → world-frame acceleration
USB Serial / Bluetooth data streaming

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Backend API (Python)
FastAPI / Uvicorn — RESTful endpoints
Signal processing & rep analysis pipeline
CSV processing and metric extraction

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Database (PostgreSQL)
Persistent storage — users, sessions, sets
Raw CSV data + computed per-rep metrics + set graph images

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Frontend (TypeScript/React)
Mantine UI — responsive dashboard
Real-time kinematic graphing & session history

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Deployment (Docker)
Docker Compose — 4 containers
Health checks, volume persistence
One-command portable deployment

Signal Processing Pipeline

1 Madgwick Orientation Filter - Orientation
On-device firmware fuses accelerometer and gyroscope data into a unit quaternion (4-tuple representation of 3D rotation) using the Madgwick algorithm. This quaternion rotates the raw acceleration vector from the sensor's local frame into world-frame and subtracts gravity, isolating the user's z-axis movement.

2 Smoothing & Dead-Zone Filter - Noise Reduction
A centered rolling average suppresses high-frequency output noise from the sensor. A dead-zone filter then zeroes out any remaining small accelerations (~ 0.6 m/s² by default) that represent sensor noise during pauses, preventing this noise from accumulating through the integration steps that follow.

3 RMS Active Region Detection - Motion Detection
A rolling Root Mean Square (RMS) aggregates the magnitude of a set of data points. When applied to the collected acceleration data, it identifies which portion of the recording contains actual barbell movement versus idle time. Only the "active" regions are processed further.

4 Double Integration + Drift Correction - Graphing and Recording
Acceleration (m/s²) is integrated over identified "active" regions to obtain velocity (m/s), then velocity is integrated again to obtain position (m). Because sensor bias causes each integration to drift, linear endpoint correction is applied - velocity is corrected such that it returns to zero at the end of the set, and position is corrected per-rep so the start and end positions match.

5 Peak-Based Rep Segmentation - Movement Delineation
To detect individual repetitions, the newly produced position curve is then searched for prominent position value peaks. The valleys between consecutive peaks define natural rep boundaries, while pauses simply widen the peak without creating a false boundary.

6 Phase Decomposition & Metrics - Final Data Processing
Finally, each rep is then split at its position peak into an eccentric phase (lowering, where muscles lengthen under load) and a concentric phase (lifting, where muscles shorten). The per-phase metrics are then calculated and stored in the database: ROM (cm), peak/avg velocity (m/s), peak/avg acceleration (m/s²), rep/rest duration (s), and average power in watts (W = mass × g × ROM / duration).

Key Metrics & Why They Matter

Range of Motion (ROM) — cm
Partial reps reduce muscle activation and increase injury risk. ROM measurement ensures each rep reaches full depth, providing objective proof of consistency across a set.

Barbell Velocity — m/s
Rep speed is the most reliable indicator of neuromuscular fatigue. When velocity drops significantly between reps, the lifter is approaching failure - a foundational principle of velocity-based training (Jovanović & Flanagan, 2014).

Rest Duration — seconds
Time paused at the top and bottom of reps is crucial for increasing muscular "time under tension," which has been shown to increase rates of muscle protein synthesis (Burd et al., 2012).

Average Power — watts
Power (W = mass × g × ROM / time) combines load, range of motion, and speed into one number - this is the best overall measure of training intensity, used in athletic performance testing and evaluation.

Features

Automated Rep Segmentation
Sets are automatically broken into individual reps with per-rep and per-phase metrics.

Rep Details

Rep	Phase	Duration	ROM	Peak Velocity	Avg Velocity	Peak Accel	Avg Accel	Avg Power	Samples
#1	ECCENTRIC ↑	0.63s	92.9 cm (0.929 m)	0.230 m/s	1.418 m/s	9.97 m/s²	6.60 m/s²	1390.7 W	43–72
	CONCENTRIC ↓	0.761s	–	0.187 m/s	1.216 m/s	7.88 m/s²	4.56 m/s²	–	72–107
#2	ECCENTRIC ↑	0.761s	88.7 cm (0.887 m)	0.216 m/s	1.147 m/s	8.38 m/s²	5.16 m/s²	1125.1 W	114–149
	CONCENTRIC ↓	0.739s	–	0.185 m/s	1.190 m/s	6.70 m/s²	4.48 m/s²	–	149–183
#3	ECCENTRIC ↑	0.739s	90.5 cm (0.905 m)	0.204 m/s	1.198 m/s	7.52 m/s²	5.07 m/s²	1174.4 W	189–223
	CONCENTRIC ↓	0.848s	–	0.173 m/s	1.061 m/s	6.55 m/s²	3.60 m/s²	–	223–262
#4	ECCENTRIC ↑	0.717s	88.6 cm (0.886 m)	0.219 m/s	1.216 m/s	9.32 m/s²	5.51 m/s²	1192.1 W	271–304
	CONCENTRIC ↓	0.826s	–	0.178 m/s	1.067 m/s	7.25 m/s²	3.98 m/s²	–	304–342
#5	ECCENTRIC ↑	0.783s	90.7 cm (0.907 m)	0.207 m/s	1.146 m/s	7.68 m/s²	4.64 m/s²	1124.2 W	349–385
	CONCENTRIC ↓	0.804s	–	0.181 m/s	1.088 m/s	6.44 m/s²	4.26 m/s²	–	385–422

Set-to-Set Comparison
Active set view highlights changes in ROM, velocity, and rest duration versus the previous set.

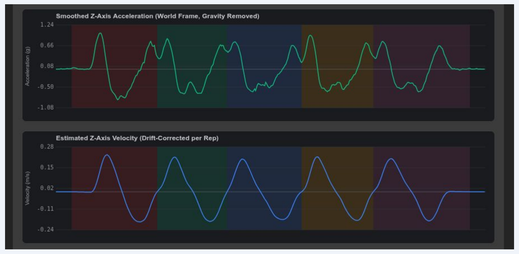
Set 1
100 kg · 5 reps
52.4 KB · Mar 11, 09:01 PM
consecutive 90cm

Avg ROM: **+0.92 cm**
Avg Rest (Top): –
Avg Rest (Bottom): –
Peak Vel Up: **+0.00 m/s**
Peak Vel Down: **+0.04 m/s**

Set Summary

Set	Weight	Reps	Avg ROM	Peak Vel Up	Peak Vel Down	Avg Rest (Top)	Avg Rest (Bottom)	Avg Power
Set 1	100 kg	5	90.3 cm	0.18 m/s	0.22 m/s	-	-	476.6 W
Set 2	100 kg	5	89.4 cm	0.18 m/s	0.17 m/s	1.40s	1.32s	212.4 W
Set 3	-	5	44.6 cm	0.11 m/s	0.12 m/s	-	0.91s	-
Set 4	10 kg	6	51.8 cm	0.11 m/s	0.11 m/s	-	0.89s	21.6 W
Set 5	-	5	48.8 cm	0.16 m/s	0.16 m/s	-	0.88s	-

Kinematic Signal Charts
Smoothed acceleration, velocity, and position traces with per-rep colored segmentation.



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Bar Brain source repository. https://github.com/EPC233/Capstone

Try It Live

Explore the web application! Currently running via Docker locally on my laptop and hosted by Ngrok.

Just click "Visit Site" and explore the app!